Study Material 3

Access the MySQL databases provided through the class database server. In the following exercises, we use the *sakila* database (a DVD rental database), which consists of 16 tables related to movie inventory, actors, customers, rental history, payment information, and more. (The database also includes several views, but you may ignore them for now.)

1. For each question below, find the answer based on the data in the database and write an SQL query that demonstrates how you obtained it. (Note: The provided query is one possible solution; other correct queries may exist.)
(a) How many stores are found in the database?

Sample Query:

Answer:

(b) How many unique last names are found in the actor relation?

Sample Query:

Answer:

(c) What is the total amount of all payments made by customer 1 (customer_id = 1)?

Sample Query:

Answer:

(d) According to the database, how many inventories (DVDs) have not been returned (inventories that have not been returned do not have return_date)?

Sample Query:

Answer:

(e) What is the average rental duration of all films in the database?

Sample Query:

Answer:

(f) How many distinct customers have rented a movie title(s) from *staff_id*=1?

Sample Query:

Answer:

(g) How many distinct films rated 'PG' are available?

Sample Query:

Answer:

(f) How many active customers are living in the district of England?

Sample Query:

Answer:

(i) How many customers live in the districts 'England', 'Texas', or 'California'?

Sample Query:

Answer:

(j) How many customers have made at least one payment?

Sample Query:

Answer:

(k) How many addresses do not have a second address line (address2)?

Sample Query:

Answer:

(l) How many films are not available in store 2?

Sample Query:

Answer:

(m) Find the films with the highest rental rate using a subquery with ALL.

Sample Query:

Answer:

In the following exercises, we use the university database, which consists of multiple tables related to students, instructors, departments, courses, classrooms, sections, enrollments, advising relationships, and prerequisite information.

2. For each question below, find the answer based on the data in the database and write an SQL query that demonstrates how you obtained it. (Note: The provided query is one possible solution; other correct queries may exist.)
(a) What is the maximum number of credits among all courses?

Sample Query:

Answer:

(b) What is the total budget of all departments?

Sample Query:

Answer:

(c) How many instructors have taught at least one section?

Sample Query:

Answer:

(d) How many students are not assigned to any advisor?

Sample Query:
```
SELECT COUNT(*)
FROM student
WHERE ID NOT IN (
   SELECT s_ID
   FROM advisor
);
```

Answer:

(e) Find the names of instructors whose salary is greater than or equal to the salary of every other instructor.

Sample Query:

Answer:

(f) Find the names of students who have taken at least one course offered by the 'Comp. Sci.' department.

Sample Query:

Answer:

(g) Find the titles of courses that are used as prerequisites for some other course.

Sample Query:

Answer:

(h) Find students whose advisor is from a different department.

Sample Query:

Answer:

(i) Find students who only take courses offered by their own department.

Sample Query:

Answer:

Note: This result is obtained using the *university* database. If the query were executed on the university_small database, it would return one or more tuples.

from *university_small*:

(j) Find instructors who have taught at least one student from the 'Comp. Sci.' department.

Sample Query:

Answer:

(k) Find students who take all courses offered by their department.

Sample Query:

Answer:

Note: This result is obtained using the *university* database. If the query were executed on the university_small database, it would return one or more tuples.

from *university_small*:

(l) Find instructors who have never taught any course

Sample Query:

Answer:

(m) Find the departments whose average instructor salary is higher than the overall average salary.

Sample Query:

Answer:

(n) Find the names of instructors whose salary is higher than the average salary of their own department.

Sample Query:

Answer:

(o) Find the instructors who teach the most sections among all instructors.

Sample Query:

Answer: